Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Projects

Rayfloat | *C++, OpenMP* | GitHub | Writeup — December 2025

- Architected a physically-based CPU renderer from scratch, implementing Lambertian, Metal, Dielectric, and DiffuseLight materials with an **iterative light transport model**, eliminating **recursion-induced stack pressure** and improving **Branch Target Buffer (BTB) predictability**.
- Diagnosed and eliminated a **41% performance penalty** from **cache-coherence contention** in `std::mt19937` by implementing a **thread-local XorShift32 pRNG**, restoring **near-linear multi-core scaling**.
- Reduced ray-scene intersection complexity from **O(n) to O(log n)** by designing a **Bounding Volume Hierarchy (BVH)** with **median-split construction** and optimized slab AABB tests, enabling scenes with 64,000+ primitives.
- Optimized **dielectric rendering** using **Schlick's Approximation** to numerically estimate the **Fresnel coefficient**, determining the probability of reflection vs refraction based on incident angles.

Spheni | *C++, Approximate Nearest Neighbor, K-Means* | GitHub — January 2026 – Present

- Architected a CPU-first vector search engine from scratch, implementing **Flat (exact)** and **IVF-Flat (ANN)** indexes with **k-means++ centroid initialization**, **inverted lists**, and **nprobe-based query routing** to expose recall–latency tradeoffs.
- Achieved a **12.6× speedup over brute-force search at ∼97% Recall@10** on 50k SIFT vectors (128D) by restricting candidate scans to nearest Voronoi partitions while maintaining tight p99 tail latency.
- Implemented heap-based **Top-K selection** with **O(N log K)** complexity and unified scoring semantics (cosine via normalized dot product, L2 via negative squared distance).
- Built a benchmark harness measuring **Recall@K**, mean/p99 latency, and throughput, identifying elbow points across **nlist/nprobe** sweeps on single-threaded CPU.

TouchGrass Chess Engine | *Python* | GitHub — October 2025

- Built a fully rules-accurate chess backend with **legal move generation**, **state transitions**, and a complete **undo/redo stack** using an object-oriented approach.
- Integrated **Zobrist hashing** for efficient board state encoding, enabling constant-time **repetition tracking** and **transposition table** lookups.
- Implemented a modular **alpha-beta search** algorithm supporting complex moves like promotions, castling, and en passant.
- Designed an **extensible API** exposing board state and termination detection, allowing integration with external UIs or engine clients.

amine | *Python, Flask, PyAutoGUI, SQLite* | GitHub | ProductHunt — August 2024 – October 2024

- Reached **#10 on ProductHunt** with **450+ downloads** by building a multi-threaded distraction-blocking productivity app.
- Implemented **system-level mouse boundary enforcement** and **keyboard interception** using PyAutoGUI to prevent application switching during focus sessions.
- Designed an analytics dashboard using **Matplotlib and Seaborn** to visualize session data with automated **CSV export** functionality.
- Engineered **SQLite-backed session management** and input control systems, gaining **170+ GitHub stars** and active community feedback.

Skills

Languages: Python, C/C++, SQL
Frameworks and Libraries: OpenMP, FastAPI, Playwright, Raylib, PyAutoGUI, pandas, NumPy, Matplotlib
Developer Tools: Git, Intel VTune Profiler, SQLite
Competitive Programming: Codeforces Pupil (1338)